



基于火灾危险气体释放与风场驱动二维对流-扩散模型的校园室外疏散方案设计方法： 以一零一中学为例

Design Method of Campus
Outdoor Fire Evacuation Plan
Based on **Hazardous Gas
Release and Wind-Driven 2D
Advection-Diffusion Model:**
A Case Study of Beijing 101 Middle School

Process Report: **Stage 2**
(Jun. Week 1)

Authors:

陈禹翔 Lambert Yuxiang Chen¹

彭博 Blake Bo Peng¹

¹Beijing 101 Middle School International Department



Chapter 1. Timeline Review 时间计划安排回顾

时间线 Timeline



北京一零一中
BEIJING 101 HIGH SCHOOL

最终提交时间 Final Deadline

本项目计划在**2026年6月第三周**完成最终递交。

The project is planned to be completed in the third week of June 2026.

第一阶段：资料整理与模型确定 | May Week 4

完成研究背景、化学原理、危险气体类型、二维对流-扩散模型和校园地图抽象方法的整理，明确研究假设与简化方案。

Collect background information, combustion chemistry, hazardous gas types, the 2D advection-diffusion model, and the simplified campus map representation.

第二阶段：模型计算与方案分析 | June Week 1

建立火源、风向、扩散范围、路径风险和集合点评价方式，比较不同情景下的疏散路线安全性。

Build fire source, wind direction, diffusion, path risk, and assembly point evaluation models, then compare evacuation safety under different scenarios.

第三阶段：论文、海报与视频制作 | June Week 2

完成论文初稿、海报主要版面和视频脚本，将化学模型、数学公式、地图图示和疏散方案整合为可展示成果。

Complete the paper draft, poster layout, and video script, integrating chemistry, mathematical formulas, campus maps, and evacuation plans.

第四阶段：修改、排版与最终提交 | June Week 3

检查逻辑完整性、双语表达、图表引用和格式规范，完成最终版本并提交论文、海报和视频。

Revise logic, bilingual wording, citations, figures, and formatting before submitting the final paper, poster, and video.



Chapter 2. Completed Content in Essay 论文完成内容

Ch2. Notation and Parameters Table (Updated)



Notations and parameters in this research are shown below.

Table 1: Notation and Parameters

Symbol	Definition	Unit / Type
Notation for Field and Discrete Field		
(x, y)	Continuous spatial location	m
(i, j)	Grid-cell index in the discrete field	–
t	Continuous time	s
n	Discrete time-step index	–
$\Delta x, \Delta y$	Grid spacing in the x and y directions	m
Δt	Time-step size	s
$\vec{w}(x, y)$	Continuous wind velocity field	m/s
$\vec{w}_{i,j}$	Wind vector at grid cell (i, j)	m/s
$u(x, y), v(x, y)$	Continuous wind components	m/s
$u_{i,j}, v_{i,j}$	Discrete wind components at cell (i, j)	m/s
$C_g(x, y, t)$	Concentration field of gas g	kg/m ³
$C_{g,i,j}^n$	Discrete concentration at cell (i, j) and time step n of gas g	kg/m ³
Geographical Information and Road Network Graph		
$G = (\mathcal{V}, \mathcal{E})$	Campus road network graph	graph
$\mathcal{V} = \{v_1, v_2, \dots, v_N\}$	Set of nodes	nodes set
$\mathcal{E} = \{e_1, e_2, \dots, e_M\}$	Set of edges	edges set
v_k	The k -th node	node
id_k	ID of node v_k	integer
(x_k, y_k)	Position of node v_k	m
$type(v_k)$	Type of node v_k	categorical
P_k	Population assigned to node v_k	persons
K_k	Capacity of node v_k , if applicable	persons
$e_m = (v_a, v_b)$	The m -th edge connecting v_a and v_b	edge
$d_m = \text{dist}(e_m)$	Distance of edge e_m	m
K_m	Capacity of edge e_m	persons
Model 1: Hazardous Gas Generation		
t	Time after ignition	s
g	Index of hazardous gas species	–
a_g	Average generation rate of gas g	kg/s
M_g	Total mass of material producing gas g	kg
$m_g(t)$	Mass of gas g generated by time t	kg
A_s	Effective source area in the 2D model	m ²
V_s	Effective source volume	m ³
$S_g(t)$	Source term of gas g	kg/(m ² s)

(Continued table)

Symbol	Definition	Unit / Type
Model 2: Wind-Driven 2D Advection-Diffusion		
$C_g(x, y, t)$	Concentration of gas g	kg/m ³
$D_{eff,g}$	Effective diffusion coefficient of gas g	m ² /s
$\vec{w}(x, y) = \langle u, v \rangle$	Wind velocity vector field	m/s
u	Vector component in x direction	m/s
v	Vector component in y direction	m/s
$S_g(x, y, t)$	Source term from Model 1	concentration/s
$S_{g,i,j}^n$	Source term in grid cell (i, j) from Model 1	concentration/s
λ_g	Attenuation coefficient of gas g	s ⁻¹
$\frac{\partial C_g}{\partial t}$	Rate of concentration change over t	concentration/s
$\frac{\partial C_g}{\partial x}$	Rate of concentration change over x	concentration/m
$\frac{\partial C_g}{\partial y}$	Rate of concentration change over y	concentration/m
$\nabla^2 C_g$	Diffusion-related spatial curvature of concentration	concentration/m ²
Model 3: Evacuation Cost and Route Optimization		
k	Index of cluster of people	–
h	Index of scheme for cluster k	–
$\mathcal{P}_k$	Set of feasible evacuation for cluster k	path set
$p_{h,k} \in \mathcal{P}_k$	A candidate full path for cluster k	path
H_g	The hazard coefficient of gas g , used in FED Method	concentration
$L(p_k)$	Total length of path p_k	m
$E(p_k)$	Exposure along path p_k	concentration · s
$C(p_k)$	Congestion level along the most crowded path p_k	dimensionless
s	The speed of crowded	m/s
$T(p_k)$	Total time consume of path p_k	s
L_0, E_0, G_0	Normalization parameters	corresponding units
α, β, γ	Weights in the cost function	dimensionless
$\text{Cost}(p_k)$	Total cost of candidate path p_k	dimensionless
p_k^*	Optimal path for cluster k	path

Ch3. Model Assumption (*Second Version*)



3 Model Assumptions

To control the scale of the study, appropriate assumptions are made for the whole model:

1. This study only discusses the outdoor evacuation process on campus after personnel have left the building. The building on fire is simplified as a source of dangerous gas release, with the release intensity estimated based on the type of burning material, burning rate, and gas generation coefficient;
2. During a short-term evacuation process, the impact of outdoor gases is primarily concentrated near the burning building, around exits, on downwind roads, and at assembly points located closer to the fire source;
3. The actual simulation employs the finite difference method to represent and calculate physical fields using grids;
4. Campus roads, building exits, and assembly points will be abstracted into a graph structure, with road segment capacity and assembly point capacity treated as known or estimable parameters in the model;
5. Evacuation behavior is handled based on population units, without further breakdown into individual complex behaviors for the time being.

3.1 Assumptions for Model 1

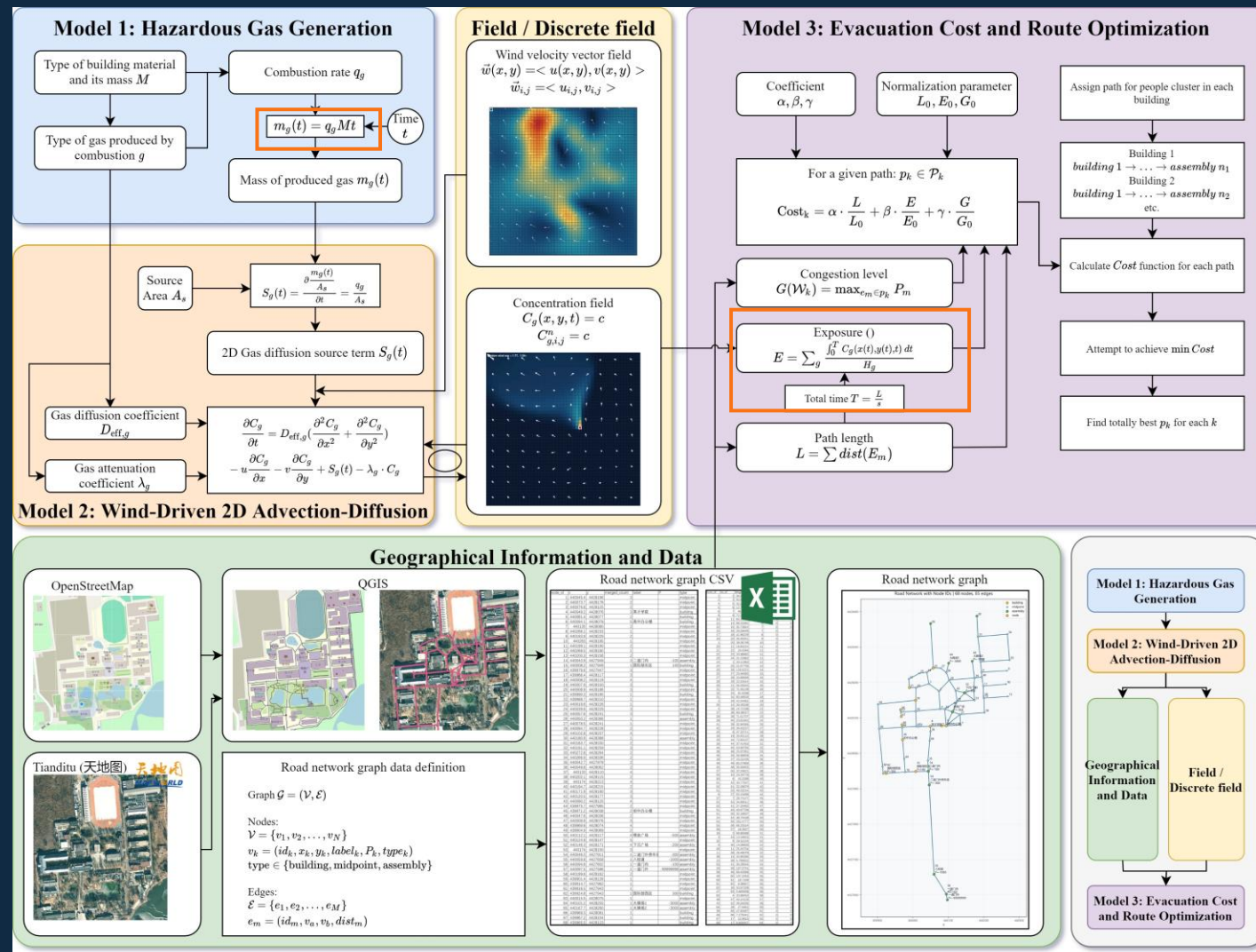
Only CO, HCN, HCl will be considered; the generation rate of each gas q_g is treated as a constant with respect to time, and the total mass of material is also treated as a constant during the short-term evacuation process.

3.2 Assumptions for Model 2

The wind field is treated as a steady-state field during the short-term evacuation process, which means that it should not be similar for all position, but should not change significantly with time.

The diffusion process is simplified to a two-dimensional model, which ignores the vertical diffusion. The effective diffusion coefficient $D_{\text{eff},g}$ is treated as a constant for each gas g .

Ch4. Model Framework (Updated)



Ch5. Data Sources and Preprocessing



5 Data Sources and Preprocessing

5.1 Geographical Map

The geographical map of the school can be obtained from geographical data supplier, such as *OpenStreetMap*, which provides detailed information about the campus layout. These data usually have three types:

1. Point: Represent some important locations, but the amount is small, such as campus entrances;
2. Segments: Represent linear features like roads, pathways;
3. Polygons: Represent area features like the outlines of buildings or green spaces.

The map data is processed to extract the necessary features for modeling. Geographical features are abstracted into a graph structure, denoted as $\mathcal{G}$ and defined as:

$$\mathcal{G} = (\mathcal{V}, \mathcal{E}) \quad (2)$$

The overall structure and data of the required graph $\mathcal{G}$ is shown below:

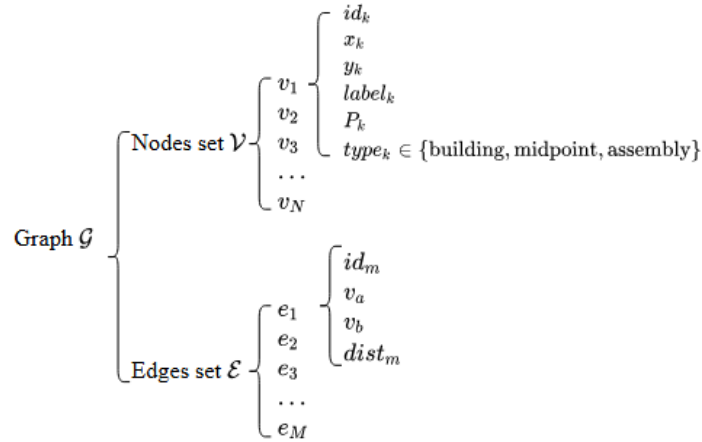


Figure 2: Geographical Map Structure Represent in Math

5.1.1 Nodes

Nodes v_k are defined to represent key locations on campus. Each one node is defined as:

$$v_k = (id_k, x_k, y_k, label_k, P_k, type_k) \quad (3)$$

- id_k : Unique identifier for node v_k , $type \in \{\text{building, midpoint, assembly}\}$;
- (x_k, y_k) : Spatial coordinates of node v_k ;
- $label_k$: Descriptive label for node v_k (e.g., "Building A Exit", "Assembly Point 1");
- P_k : Population assigned to node v_k (if applicable);
- $type_k$: Categorical variable indicating the type of node (e.g., "building", "midpoint", "assembly").

According to the map data, the main types of nodes include building nodes, road midpoints, and assembly points.

Building nodes

As for the building nodes, which are the starting points of evacuation, they are located by the main entrances of each crowded building. Most of time, only one main entrance is considered for each building, while for some large buildings with multiple entrances, more than one node can be defined to represent the different exits.

The population P_k assigned to each building node can be estimated based on the building's function and typical occupancy, such as classrooms, dormitories, or administrative offices. If the building has multiple exits, the population can be distributed among the corresponding nodes based on the estimated usage of each exit.

Road midpoints

Road midpoints are defined as nodes that represent the cross points of road segments. They are used to model the connectivity and flow of people along the road network.

Assembly points

Assembly points are defined as nodes that represent designated safe areas where evacuees can gather after evacuating from buildings. These points are choose manually and typically located in open spaces and are designed to accommodate a certain number of people. For example, plaza, sports field, or parking lot can be used as assembly points.

The population P_k of each assembly point can be estimated based on the area of the location and safety guidelines for crowd density. This capacity is important for modeling the evacuation process, as it can influence the routing decisions and congestion levels. Notice that the capacity P_k is negative for assembly points, which means that they can receive people, while the building nodes have positive P_k which means they are the source of people.

5.1.2 Edges

Edges e_m represent the walkable paths between nodes, which are obtain and simplified from the road segments in the map data. Each edge is defined as:

$$e_m = (id_m, v_a, v_b, dist_m) \quad (4)$$

- id_m : Unique identifier for edge e_m ;
- v_a, v_b : The two nodes that edge e_m connects, which can be passed by two direction;
- $dist_m$: The distance of edge e_m , which can be calculated based on the spatial coordinates of the connected nodes.

5.2 Wind Field

The wind field $\vec{w}(x, y) = \langle u(x, y), v(x, y) \rangle$ is obtained from local meteorological data, which provides information about the wind speed and direction at different locations on campus. The continuous wind field is discretized into a grid-based representation $\vec{w}_{i,j} = \langle u_{i,j}, v_{i,j} \rangle$ that corresponds to the grid used in the diffusion model.

5.2.1 Simplified Wind Field

To simplify the model, the wind field can be assume that is uniform across the campus, which means that the wind speed and direction are the same at all locations. In this case, the wind field can be represented as a constant vector $\vec{w} = \langle u, v \rangle$, where u is the wind speed in the x direction and v is the wind speed in the y direction, and $\langle u, v \rangle$ is similar for all x, y . This method also able to apply on discrete grid, which means that the wind vector $\vec{w}_{i,j}$ is the same for all grid cells (i, j) .

This simplification allows us to focus on the overall impact of wind on gas diffusion without needing to model complex spatial variations in the wind field.

5.2.2 Standard Wind Field

Alternatively, a more detailed wind field can be constructed based on the meteorological data, which captures the spatial variations in wind speed and direction across the campus. This can be done by interpolating the wind data from meteorological stations or using computational fluid dynamics (CFD) simulations to model the wind flow around buildings and other structures. The resulting wind field $\vec{w}(x, y)$ can then be discretized into the grid-based representation $\vec{w}_{i,j}$ for use in the diffusion model. This approach provides a more accurate representation of the wind's influence on gas diffusion, but it also increases the complexity of the model.

6 Model 1: Hazardous Gas Generation Model

This model is aim to quantify the amount of gas released during the combustion of teaching buildings in schools, verus to time t .

Ch6. Model 1: Hazardous Gas Generation Model



6.1 Building Material M_b and Its Combustion Product

While the hazardous gases generated by combustion can vary widely based on the specific materials involved, we focus on building materials found in school buildings which are the most common but harmful after its combustion.

In the common school buildings arrangement, the main flammable building materials include Polyurethane foam (PU), Polyvinyl chloride (PVC), and Polystyrene (PS) etc, and the combustion of these materials can produce a variety of hazardous gases, including smoke partite, carbon monoxide (CO), hydrogen cyanide (HCN) etc.

Table 2 below shows the main building materials and their corresponding hazardous gas products.

Table 2: Common combustible materials in school buildings and their main products

Material (Formula)	Typical Locations	Combustion Products
Polyvinyl Chloride (PVC) ^{1,2}	Electrical cables, vinyl flooring, pipes	CO(g), HCl(g)
	$(C_2H_3Cl)_n + O_2(g) \rightarrow CO(g) + HCl(g) + H_2O(g) + CO_2(g)$	
Polyurethane Foam (PU) ³	Chair cushions, sofas, insulation	CO(g), HCN(g), NO _x (g)
	$PU + O_2(g) \rightarrow CO(g) + HCN(g) + CO_2(g) + H_2O(g) + NO_x(g)$	
Polystyrene (PS) ⁴	Packaging, insulation boards	Styrene(g), CO(g)
	$(C_8H_8)_n + O_2(g) \rightarrow C_8H_8(g) + CO(g) + CO_2(g)$	
Polyethylene (PE) ⁴	Cable jackets, containers, plastic products	CO(g), Hydrocarbons(g)
	$(C_2H_4)_n + O_2(g) \rightarrow CO(g) + CO_2(g) + H_2O(g)$	
Polypropylene (PP) ⁴	Chairs, storage boxes, laboratory equipment	CO(g), Hydrocarbons(g)
	$(C_3H_6)_n + O_2(g) \rightarrow CO(g) + CO_2(g) + H_2O(g)$	
Recycled Board ⁴	Desks, cabinets, partitions	CO(g), HCHO(g), PM(s)
	$(C_6H_{10}O_5)_n + O_2(g) \rightarrow CO(g) + CO_2(g) + HCHO(g) + H_2O(g)$	
Synthetic Rubber (SBR) ⁴	Flooring, mats, sealing strips	CO(g), SO _x (g)
	$SBR + O_2(g) \rightarrow CO(g) + CO_2(g) + SO_x(g) + H_2O(g)$	
Acrylonitrile Butadiene Styrene (ABS) ⁴	Computer cases, laboratory devices	CO(g), Acrylonitrile(g)
	$ABS + O_2(g) \rightarrow CO(g) + C_3H_3N(g) + CO_2(g) + H_2O(g)$	

¹ Doe (1987), *The Combustion Toxicology of Polyvinylchloride Revisited*.

² Huggett and Levin (1986), *Toxicity of the Pyrolysis and Combustion Products of Poly(Vinyl Chlorides)*.

³ McKenna and Hull (2016), *The Fire Toxicity of Polyurethane Foams*.

⁴ Stec and Hull (2011), *Assessment of the Fire Toxicity of Building Insulation Materials*.

Although the actual combustion process can be complex and produce many different

gases, this table summarizes the main hazardous gases that are commonly produced by the combustion of typical school building materials. Also, most of the products above are hazardous gases, but some of them can be diluted to non-hazardous levels in outdoor environments quickly. Thus, In this study, the model will be simplified to focus on a few key hazardous gases, which are **CO, HCN, and HCl**, which are known to be particularly harmful in fire situations, while diffused to outdoors.

6.2 Generation Rate q_g of gas g

After identifying the main hazardous gases, the next step is to estimate the generation rate q_g of each gas g . The generation rate of each gas can be estimated as a average constant value with respect to the mass M_g of the material that produces gas g , denoted as q_g , and has unit of $kg \cdot s^{-1} \cdot kg^{-1}$.

The mass of material decreases slowly and slightly if compare with its total mass, so M_g can be treated as a constant during the short-term evacuation process. Therefore, the generation rate of gas g can be expressed as:

$$\frac{dm_g(t)}{dt} = q_g M \quad (5)$$

Thus, the total mass of gas $g(t)$ respect to time t can be calculated as:

$$m_g(t) = \int_0^t q_g M dt = q_g M t \quad (6)$$

Which estimate the total mass of gas g generated by time t as a linear function of time and mass, with the slope being the generation rate q_g .

The values of q_g for chosen gas are shown below:

Table 3: Typical Generation Rates of Key Hazardous Gases

Gas	Generation Rate q_g (kg/s/kg) ¹		
	MIN.	MAX.	TYP.
Carbon Monoxide (CO)	2.0×10^{-4}	2.0×10^{-3}	1.0×10^{-3}
Hydrogen Cyanide (HCN)	5.0×10^{-5}	5.0×10^{-4}	5.0×10^{-4}
Hydrogen Chloride (HCl)	2.0×10^{-3}	8.0×10^{-3}	4.0×10^{-3}

¹ (Purser & McAllister, 2016), *Assessment of Hazards to Occupants from Smoke, Toxic Gases, and Heat*.

Ch7. Model 2: Hazardous Gas Generation Model



7.1 Source Term Construction

The source term $S_g(t)$ will enter the diffusion model as the generation of gas g from the fire, which is calculated based on the mass function $m_g(t)$ from Model 1. The source term $S_g(t)$ represents the rate of gas generation with the source area A_s , which can be calculated as the time derivative of the mass function divided by the effective source area A_s :

$$S_g(t) = \frac{\partial \frac{m_g(t)}{A_s}}{\partial t} = \frac{\dot{q}_g}{A_s} \quad (7)$$

For continuous field, $S_g(t)$ is treated as a point source at the location of the fire, which means that the gas is generated at a specific point in space. For discrete grid, $S_{g,i,j}$ is treated as a source term in the grid cell (i, j) that contains the fire, which means that the gas is generated within that grid cell.

7.2 Fick's Law and its Expression in Advection-Diffusion Equation

Fick's law is the foundation of the diffusion model, which states that the flux of a substance is proportional to the negative gradient of its concentration. In the context of hazardous gas diffusion, Fick's law can be expressed as:

$$\vec{J}_g = -D_{\text{eff},g} \nabla C_g \quad (8)$$

In this case, because it happens in 2D outdoor environment and the effect of wind is significant, so the diffusion process is modeled as a two-dimensional advection-diffusion process, which can be described by the following partial differential equation:

$$\frac{\partial C_g}{\partial t} = D_{\text{eff},g} \left(\frac{\partial^2 C_g}{\partial x^2} + \frac{\partial^2 C_g}{\partial y^2} \right) - u \frac{\partial C_g}{\partial x} - v \frac{\partial C_g}{\partial y} + S_g(t) - \lambda_g \cdot C_g \quad (9)$$

- $\frac{\partial C_g}{\partial t}$: The rate of change of concentration of gas g with respect to time t .
- $D_{\text{eff},g} \left(\frac{\partial^2 C_g}{\partial x^2} + \frac{\partial^2 C_g}{\partial y^2} \right)$: The diffusion term, which represents the spreading of gas g due to concentration gradients. $D_{\text{eff},g}$ is the effective diffusion coefficient for gas g .
- $-u \frac{\partial C_g}{\partial x} - v \frac{\partial C_g}{\partial y}$: The advection term, which represents the transport of gas g due to the wind field. u and v are the wind velocity components in the x and y directions, respectively.
- $S_g(t)$: The source term, which represents the generation of gas g from the fire, as calculated in Model 1.
- $-\lambda_g \cdot C_g$: The attenuation term, which represents the natural decay or removal of gas g over time, where λ_g is the attenuation coefficient for gas g .

As for discretization in finite difference methods, the equation can be approximated as:

$$\begin{aligned} C_g^{n+1}(i, j) = & C_g^n(i, j) \\ & + \Delta t \left[D_{\text{eff},g} \left(\frac{C_g^n(i+1, j) - 2C_g^n(i, j) + C_g^n(i-1, j)}{\Delta x^2} \right. \right. \\ & \left. \left. + \frac{C_g^n(i, j+1) - 2C_g^n(i, j) + C_g^n(i, j-1)}{\Delta y^2} \right) \right. \\ & \left. - u \frac{C_g^n(i+1, j) - C_g^n(i-1, j)}{2\Delta x} - v \frac{C_g^n(i, j+1) - C_g^n(i, j-1)}{2\Delta y} \right. \\ & \left. + S_g^n(i, j) - \lambda_g C_g^n(i, j) \right] \end{aligned} \quad (10)$$

- $C_g^n(i, j)$ and $C_g^{n+1}(i, j)$ denote the concentration of gas g at grid cell (i, j) before and after one time step, respectively.
- The terms multiplied by $D_{\text{eff},g}$ represent molecular and turbulent diffusion in both spatial directions.
- The terms containing u and v describe advection caused by wind transport.
- $S_g^n(i, j)$ represents the gas generation source from the fire.
- $\lambda_g C_g^n(i, j)$ accounts for gas removal due to ventilation, deposition, and chemical decay.
- Δt , Δx , and Δy are the temporal and spatial discretization intervals.

7.3



Chapter 3. Completed Content for Simulation 模拟程序完成内容

Program



北京一零一中
BEIJING 101 HIGH SCHOOL

技术栈: Python, PyQt前端 + 自研后端。支持CPU串行/CPU并行/GPU CUDA计算。

Technology stack: Python, PyQt frontend + self-developed backend. Supports CPU serial/CPU parallel/GPU CUDA computing.

```
app.py x core.py 8 {} environment.json
app.py > run_self_test
4604 def run_cache_self_test() -> None:
4646     print(
4647         "cache-self-test ok: "
4648         f"cache={cache.dir}, snapshots={snapshot_count}, "
4649         f"status={status.get('state', 'n/a')}, pixmap_cached={first_pixmap is not None}"
4650     )
4651
4652
4653 def main() -> None:
4654     if "--self-test" in sys.argv:
4655         run_self_test()
4656         return
4657     if "--cache-self-test" in sys.argv:
4658         run_cache_self_test()
4659         return
4660     if "--render-preview" in sys.argv:
4661         index = sys.argv.index("--render-preview")
4662         target = Path(sys.argv[index + 1]) if index + 1 < len(sys.argv) else APP_DIR / "temp" / "pre"
4663         target.parent.mkdir(parents=True, exist_ok=True)
4664         render_preview(target)
4665         return
4666     app = QApplication(sys.argv)
4667     configure_fonts()
4668     window = MainWindow()
4669     window.show()
4670     sys.exit(app.exec())
4671
4672
4673 if __name__ == "__main__":
4674     main()
4675
```

```
app.py core.py 8 x {} environment.json
core.py > ...
1844
1845
1846 def app_relative(path: Path) -> str:
1847     try:
1848         return str(path.resolve().relative_to(APP_DIR).as_posix())
1849     except ValueError:
1850         try:
1851             return "../" + str(path.resolve().relative_to(ROOT_DIR).as_posix())
1852         except ValueError:
1853             return str(path.resolve())
1854
1855
1856 def deep_merge(base: dict[str, Any], update: dict[str, Any]) -> dict[str, Any]:
1857     result = dict(base)
1858     for key, value in update.items():
1859         if isinstance(value, dict) and isinstance(result.get(key), dict):
1860             result[key] = deep_merge(result[key], value)
1861         else:
1862             result[key] = value
1863     return result
1864
1865
1866 def clamp(value: int, lower: int, upper: int) -> int:
1867     return max(lower, min(upper, value))
1868
```

● default分组限时0.01x倍率, 6.6号18:00 - 6.7号23:59
1 天前 2026-06-06 18:06

正常运行时间
4:07:42:21

Features

基础设置

T max (s) 200

dt (s) 0.0467974

dx (m) 7

dy (m) 7

颜色上限 0.123

地理地图

Nodes: ./QGIS project/qgis-osm/nodes_raw_snapped.csv

Edges: ./QGIS project/qgis-osm/edges_raw_snapped.csv

导入 Edges

导入 Nodes

风场

manual | avg <5.14, 5.36> m/s | grid 115 x 54

打开风场编辑器

Model 1: 气体生成

气体集预设 Fire toxicity

气体预设 自定义/默认

自定义名称

添加集合

添加气体

启用	气体	q_g kg/(s kg)	M_g kg	A_g m ²	$D_{eff,g}$ m ² /s	λ_g s ⁻¹	删除
☒	CO	1	100	25	0.12	0.001	删除
色带							
☒	HCN	1	80	25	0.1	0.0015	删除
色带							
☒	HCI	0.004	60	25	0.09	0.02	删除
色带							

Model 2: 扩散方程

源点节点 3: midpoint (midpoint)

方程 $S_g = q_g M_g / A_g$, $\partial C / \partial t = D^2 C - u \partial C / \partial x - v \partial C / \partial y + S - \lambda C$

Model 3: 路径寻路

算法 Dijkstra: 最短路径

起点 所有 building 节点

终点节点 62: 国际部西区 (building, P=300)

终点 综合代价最低集合点

终点节点 14: 二道门内 (assembly, P=100)

风险气体 所有启用气体

s (m/s) 1.2

人数 100

K_r m 250

alpha 1

beta 100

gamma 0.35

L0 (m) 0

E0 1

G0 1

空间采样 (m) 0

暴露积分 dt (s) 2

☒ 显示路线 ☒ 自动更新

计算路线

清除路线

已从 t=0 计算 11 条 building 路线。

总 L=1276.4 m, 最大 E=6.754

路线: S: building -> 50: 操场广场

S: building -> 50: 操场广场 | L=55.9 m | T=46.6 s | E=1.47e-23 | Cost=55.9

6: 高中办公楼 -> 50: 操场广场 | L=43.2 m | T=36.0 s | E=1.35e-29 | Cost=43.2

15: 国际部西区 -> 14: 二道门内 | L=35.7 m | T=29.8 s | E=6.75 | Cost=35.7

19: building -> 50: 操场广场 | L=136.3 m | T=113.6 s | E=1.05e-38 | Cost=136

21: building -> 64: 大操场1 | L=170.6 m | T=142.2 s | E=0 | Cost=171

25: building -> 64: 大操场1 | L=69.3 m | T=57.7 s | E=0 | Cost=69.3

45: 高中办公楼 -> 14: 二道门内 | L=156.4 m | T=130.4 s | E=6.75 | Cost=156

62: 国际部西区 -> 14: 二道门内 | L=119.3 m | T=99.4 s | E=6.75 | Cost=119

66: building -> 50: 操场广场 | L=175.0 m | T=145.8 s | E=1.8e-22 | Cost=175

67: building -> 50: 操场广场 | L=160.3 m | T=131.6 s | E=1.05e-38 | Cost=160

68: building -> 50: 操场广场 | L=154.2 m | T=128.5 s | E=1.05e-38 | Cost=154

校园气体扩散工作台

文件(F) 编辑(E) 模拟(S) 可视化(V) 帮助(H)

基础设置

T max (s) 200

dt (s) 0.0467974

dx (m) 7

dy (m) 7

颜色上限 0.123

地理地图

Nodes: ./QGIS project/qgis-osm/nodes_raw_snapped.csv

Edges: ./QGIS project/qgis-osm/edges_raw_snapped.csv

导入 Edges

导入 Nodes

风场

manual | avg <5.14, 5.36> m/s | grid 115 x 54

打开风场编辑器

Model 1: 气体生成

气体集预设 Fire toxicity

气体预设 自定义/默认

自定义名称

添加集合

添加气体

启用	气体	q_g kg/(s kg)	M_g kg	A_g m ²	$D_{eff,g}$ m ² /s	λ_g s ⁻¹	删除
☒	CO	1	100	25	0.12	0.001	删除
色带							
☒	HCN	1	80	25	0.1	0.0015	删除
色带							
☒	HCI	0.004	60	25	0.09	0.02	删除
色带							

模拟

重置

渲染

播放

单步

气体色标

CO

HCN

HCI

气体图层

气体 显示 透明度

CO ☒ 100%

HCN ☐ 100%

HCI ☒ 43%

manual wind avg <5.14, 5.36> | grid 115 x 54

大操场1 大操场2

下式广场

顺泉广场

英才中学中办公楼

初中办公楼

国际部西区 国际部东部门内

二道门外停车场

入校道

一道门内 一道门外

t = 0.00 s



北京一零一中
BEIJING 101 HIGH SCHOOL

Thanks!

Process Report: Stage 2 (Jun. Week 1)

Authors:

陈禹翔 Lambert Yuxiang Chen¹

彭博 Blake Bo Peng¹

¹Beijing 101 Middle School International Department